

A Major Project Phase-II Report
On
**"A Novel Amphibious Robot with Transformable
Leg Flipper Composite Propulsion Mechanism"**

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IN
ELECTRONICS & COMMUNICATION ENGINEERING

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Certificate

This is to certify that the Major Project Phase-II report entitled “**A Novel Amphibious Robot with Transformable Leg-flipper Composite Propulsion Mechanism**” being submitted by **Ms.T.HARIPRIYA**(1608-20-735-001), **Mr.K.KARTHIK**(1608-20-735-034),**Ms.CH.SREE RAMYA** (1608-20-735-041) in partial fulfillment for the award of the Degree of Bachelor of Engineering in Electronics and Communication Engineering of the Osmania University, Hyderabad, during 2023-24, is a record of bonafide work carried out under our guidance and supervision.

The results presented in this project report have not been submitted to any other university or institute for the award of any Degree or Diploma.

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DECLARATION

This is to declare that the work submitted in the present project work report titled **“A Novel Amphibious Robot with Transformable Leg-flipper Composite Propulsion Mechanism”** is a record of bonafide work done by us in the Department of Electronics & Communication Engineering, Matrusri Engineering College, Saidabad, Hyderabad.

No part of the report is copied from books, journals, internet and wherever the subject content is taken, the same has been duly referred in the text. The report generated is based on the project work carried out entirely by us and not copied from any other source.

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We would like to take this opportunity to place it on the record, that this project would never have taken shape but for the cooperation extended to us by certain individuals. Though this is not possible to name all of them, it would be a pardon on our part if we don't mention some of the very important persons Sincerely, we acknowledge our deep sense of gratitude to the project guide, **Dr. B. Indira Priyadarshini**, Assistant Professor for her constant encouragement, help and valuable suggestions. We wish to thank her for her constant motivation and help throughout the project.

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ABSTRACT

The allure and complexity of the amphibious robot lie in its versatile applications and the intricate challenges posed by its operational environments. Navigating rough terrain and seamlessly transitioning through various landscapes, including sand and mud, make it a compelling technological endeavor.

In response to such demands, our study introduces the innovative amphibious robot featuring a transformative leg-flipper composite propulsion system. Our project delves into the intricacies of the design for the transformative leg-flipper propulsion mechanism and its corresponding drive module. These components empower the amphibious robot to navigate through diverse terrains, seamlessly transitioning between land, water, and areas in between. The structural design details are thoroughly presented, emphasizing the crucial role played by the transformable leg-flipper propulsion in facilitating the robot's movement across different environments. To better understand the interaction dynamics between the elliptic leg and transitional environments, particularly granular mediums, a preliminary theoretical analysis is conducted. The focus is on comprehending how the robot's leg interacts with terrains, shedding light on its adaptability and efficiency in diverse conditions.

An orthogonal experiment is meticulously designed to investigate the leg's locomotion in sandy and muddy terrains, considering variations in water content. The culmination of these efforts is seen in the basic propulsion experiments conducted on the robot. These experiments serve to validate the transformative leg-flipper mechanism's efficacy in achieving locomotion capabilities both on land and underwater. The results affirm the successful implementation of the proposed design, showcasing the robot's ability to navigate various terrains with the versatile transformable leg-flipper propulsion system.

Keywords: Amphibious robot, Drive module, Leg Flipper Composite Propulsion System, Elliptic Leg, Locomotion

CONTENTS

	Page No
Certificate	i
Declaration	ii
Acknowledgement	iii
Abstract	iv
List of Figures	vii
Chapter 1: INTRODUCTION	1
1.1 Introduction	1
1.2 Problem Statement	2
1.3 Objective	2
1.4 Specifications	2
1.5 Methodologies	2
1.6 Motivation	3
1.7 Thesis Organization	4
Chapter 2: Literature Survey	6
Chapter 3: Problem Specification	11
3.1 Existing System	11
3.2 Block Diagram	12
3.3 Requirements	16
Chapter 4: Hardware Implementation	18
4.1 Introduction	18
4.1.1 Mechanical Design	18
4.1.2 Electronical Design	20
4.2 Design Description	21
4.2.1 Mechanical Components	21
4.2.2 Electronic Components	28
Chapter 5: Software Implementation	35
5.1 Arduino IDE	35

5.2 Blynk Application	37
5.3 RCFPV PRO Application	39
Chapter 6: Integration and Testing	41
6.1 Integrating the Esp 8266 & Arduino IDE	41
6.2 Testing	41
Chapter 7: Results	44
7.1 Robot	44
7.2 Results	44
Chapter 8: Conclusion	47
8.1 Conclusion	47
8.2 Future Scope	47
References	49
Appendix	50

LIST OF FIGURES

Figure No.	Name of Figure	Page No.
Fig 3.1	Block Diagram	13
Fig 3.2	Molded plastic links	15
Fig 4.1	Acrylic Molded Head Outer Parts	21
Fig 4.2	Hallow Plastic Rear Motor Housing	23
Fig 4.3	Copper Helix	24
Fig 4.4	3D Printed Curved Spoke Wheels	24
Fig 4.5	Plastic Links	28
Fig 4.6	ESP 8266	29
Fig 4.7	Camera Module	30
Fig 4.8	L298 Motor Driver	31
Fig 4.9	100 RPM DC Motor	32
Fig 4.10	Rain Sensor	34
Fig 5.1	Arduino IDE	35
Fig 5.2	Blynk Application	37
Fig 5.3	RCFPV PRO Application	39
Fig 6.1	Programming for Arduino IDE	43
Fig 6.2	Interfacing Arduino IDE	43
Fig 7.1	The Robot Movement on Water	45
Fig 7.2	The Robot Movement on Land	46

Chapter 1

INTRODUCTION

1.1 Introduction

The development of novel robotic systems has seen remarkable advancements in recent years, with an increasing focus on versatile locomotion mechanisms that can navigate diverse environments.

One such innovation is the wave-like crawling robot, designed specifically to excel in both terrestrial and aquatic settings. This project introduces the concept and design of this unique robot, highlighting its exceptional swimming capabilities and its potential applications in various fields.

The wave-like crawling robot draws inspiration from nature, mimicking the undulating motion of snakes and eels to propel itself through different mediums. This biomimetic approach not only enhances its efficiency but also allows for seamless transition between crawling on land and swimming in water. The robot's flexible body, composed of interconnected segments, enables it to navigate complex terrain with ease while maintaining stability and agility.

One of the key features of this robot is its ability to adapt its locomotion strategy based on environmental conditions. When faced with obstacles or uneven surfaces, it can adjust the frequency and amplitude of its waves to overcome challenges effectively. Similarly, in aquatic environments, it utilizes its wave-like motion to generate propulsion, enabling swift and efficient swimming.

The wave-like crawling robot's exceptional swimming capabilities open up a wide range of potential applications across various industries. In search and rescue operations, it can navigate through debris and navigate confined spaces to locate and assist survivors. In environmental monitoring, it can explore underwater ecosystems with precision, collecting data on water quality and marine life. Additionally, it holds promise in infrastructure inspection, pipeline maintenance, and underwater exploration, where traditional robots struggle to operate effectively.

In summary, the wave-like crawling robot represents a significant advancement in robotic locomotion, offering unparalleled versatility and efficiency in both terrestrial and aquatic environments. Through its innovative design and exceptional capabilities, it promises to revolutionize various fields and pave the way for new developments in robotics and automation.

1.2 Problem Statement

Contemporary robotics struggle with adaptability, hindering seamless transitions between land and water due to inefficient propulsion. To address this, our project designs an amphibious robot with a transformative leg-flipper propulsion system, enhancing versatility for disaster response, environmental monitoring, and exploration.

1.3 Objective

To develop an amphibious robot equipped with a transformative leg-flipper composite propulsion system. The overarching goal is to demonstrate the robot's seamless transition between land and water, making it suitable for application such as disaster response, search and rescue missions, environmental monitoring. Through comprehensive propulsion experiments, we aim to verify the efficiency, stability, and adaptability of the transformative leg-flipper mechanism. Ultimately, this project contributes to the advancement of robotics and autonomous systems, with a commitment to documenting and sharing knowledge within the scientific and robotics communities.

1.4 Specifications

Specific requirements are Wave-like crawling mechanism for versatile terrain navigation. To develop a versatile amphibious vehicle capable of navigating both land and water with equal efficiency. Its design incorporates advanced propulsion systems, including both wheels and helix, enabling seamless transitions between terrains. Equipped with robust suspension and buoyancy control mechanisms, the Amphibious robot ensures stability and safety in varied environments. Its modular architecture allows for easy customization to suit diverse applications, from military operations to civilian transport and exploration. With emphasis on adaptability and performance, the robot sets a new standard in amphibious vehicle technology, promising enhanced mobility and functionality in challenging conditions. Advanced swimming capabilities with integrated propulsion systems. Robust construction with waterproof materials, Sensors, navigation, autonomous control, and rechargeable power source.

1.5 Methodologies

The methodology involves initial research on wave-like crawling and swimming mechanisms followed by prototyping and testing of selected designs, iterating through optimization phases. The first step of this project is to dive deep into researching wave-like crawling and swimming mechanisms. We will research into existing literature and

cutting-edge technologies to find effective principles and innovative approaches. These insights will then guide us in conceptualizing and designing initial prototypes.

Once the research phase is completed, we will move to prototyping and testing our chosen designs. This process involves creating preliminary models and subjecting them to rigorous tests. Our objective is to evaluate their performance, identify any areas that can be improved, and refine the designs accordingly. Each iteration is aimed at making the prototypes more efficient, reliable, and functional.

As our designs progress, we'll focus on integrating different components into a cohesive system. This phase is crucial because it involves developing and implementing advanced control algorithms. These algorithms are designed to ensure that the system can operate autonomously, performing complex tasks with minimal human intervention.

To evaluate the effectiveness of the integrated system, we will conduct comprehensive field tests. These tests will be carried out in environments that closely resemble real-world conditions, providing us with valuable data on how the system performs. We'll measure and analyze parameters such as speed, stability, energy consumption, and adaptability in meticulous detail.

The findings from these field tests will be documented extensively, laying the foundation for further refinements. This documentation will serve as a roadmap for future iterations, highlighting areas where improvements can be made and suggesting potential modifications. Our ultimate goal is to create a highly optimized and autonomous system that can perform efficiently in its intended applications.

Throughout this methodology, we'll maintain a formal yet humanized approach. We will ensure that each phase is conducted with precision while also considering the broader implications and potential applications of the technology. This balanced perspective will help us develop innovative solutions that are both technically sound and practical.

The aim of combining system and user prompts is to optimize the assistant's ability to refine the text into a more human-like version while strictly adhering to the original content's intent and factual correctness.

1.6 Motivation

Amphibious robot is driven by the recognition of the limitations posed by traditional vehicles when it comes to navigating diverse terrains. Conventional land vehicles excel on roads and solid ground, while helix and links are efficient on water. However, the transition between these environments often presents challenges, requiring different

modes of transportation and infrastructure. This limitation can be particularly pronounced in scenarios such as military operations, disaster response, exploration missions, or even everyday transportation in regions with complex geography.

The motivation for our project lies in addressing these challenges comprehensively. By developing an innovative, all-terrain vehicle capable of seamlessly transitioning between land and water, the project aims to offer a singular solution to diverse mobility needs. This vehicle's design integrates advanced propulsion systems, such as wheels for land travel and Helix with links for aquatic movement, ensuring smooth transitions between terrains without compromising performance or safety.

Furthermore, the project recognizes the importance of adaptability and versatility in meeting various operational requirements. The modular architecture of the project enables customization to suit specific applications, whether for military purposes, civilian transportation, or scientific exploration. This adaptability extends to factors such as payload capacity, communication systems, and onboard equipment, ensuring that the vehicle can fulfill a wide range of roles effectively.

Ultimately, the project is to revolutionize transportation by providing a single, multifunctional platform capable of overcoming the challenges posed by diverse terrains. By offering enhanced mobility and accessibility, the project aims to improve operational effectiveness across different sectors while paving the way for innovative solutions to complex transportation needs.

1.7 Thesis Organization

Chapter 1 serves as an introductory section, providing a comprehensive overview of the thesis and the project at hand. Within the project overview, a succinct introduction to the project is offered, alongside a delineation of the problem statement, objectives, methodologies, and the driving motivation behind the endeavor. Chapter 2 is dedicated to an extensive literature survey on the topic. Here, a thorough examination of prior scholarly works related to the project is conducted, with a particular focus on elucidating their limitations and shortcomings. Chapter 3 is devoted to the meticulous specification of the problem. This entails a detailed elucidation of the project's objectives, a comprehensive block diagram delineation, an exposition of the hardware components involved, and an elucidation of how the design is translated into hardware and software components. Moving forward, Chapter 4 meticulously details the hardware implementation aspects of the project, offering a comprehensive insight into the technical intricacies involved. Chapter 5 proceeds to expound upon the project's

description, meticulously delineating each module's interface and its subsequent integration into the software implementation. In Chapter 6, the focus shifts towards the critical phases of integration and testing, elucidating the methodologies employed and the outcomes garnered therein. Chapter 7 serves as the platform for presenting the results derived from the final output, providing stakeholders with a comprehensive understanding of the project's efficacy and performance. Finally, Chapter 8 offers a cogent conclusion to the project, supplemented by insightful reflections on potential future avenues for exploration and expansion.

Chapter 2

LITERATURE SURVEY

Yucheng Tang, Xiaolong Yang, Xiaojin Zhu, Shichao Zhou, Wenbin Zha, Yuxin Sun, and Yulin Wang. A 2 degrees of freedom united propulsive mechanism for amphibious function inspired by frog's hindlimb.

The methodology involves a biomechanical analysis of frog hindlimbs to understand their propulsion mechanisms, followed by the development of conceptual designs for a 2-degree-of-freedom united propulsive mechanism inspired by these limbs. Prototypes are then built and integrated with an amphibious robotic platform for testing in various environments. Through extensive testing and optimization, control algorithms are developed to coordinate movement and ensure smooth transitions between land and water locomotion modes. Finally, the performance of the united propulsive mechanism is evaluated and compared with existing systems to assess its effectiveness in achieving amphibious function.

Mohammed Rafeeq, Siti Fauziah Toha, Salmiah Ahmad and Mohd Asyraf Razib. Locomotion Strategies for Amphibious Robots-A Review.

The methodology for a review article on locomotion strategies for amphibious robots involves several key steps. Firstly, the research scope is clearly defined, specifying the types of amphibious robots and locomotion strategies to be covered. A comprehensive literature search is then conducted using relevant academic databases and other sources. Inclusion and exclusion criteria are established to screen the search results, and selected articles are thoroughly reviewed to extract pertinent information such as locomotion methods, design principles, and performance metrics. The findings are synthesized into thematic categories, allowing for a comparison of different strategies and their respective advantages and limitations. Through analysis and discussion, the effectiveness of various locomotion strategies in different environments is evaluated along with potential applications and areas for future research. The review concludes by summarizing key findings and identifying gaps or limitations in the existing research, with the final manuscript undergoing revision and finalization before submission or publication.

Robert Baines, Simon Freeman, Frank Fish³ and Rebecca Kramer- Bottiglio. Variable stiffness morphing limb for amphibious legged robots inspired by chelonian environmental adaptations.

The methodology for developing a variable stiffness morphing limb for amphibious

legged robots, inspired by chelonian environmental adaptations, likely involves a multidisciplinary approach merging biology, robotics, and materials science. Initially, a thorough investigation of chelonian (turtle and tortoise) limb morphology and biomechanics would be conducted to identify key features contributing to their adaptability in terrestrial and aquatic environments. This would involve literature review, observation, and possibly experimentation with live chelonians. Subsequently, based on the insights gained, a conceptual design for the robotic limb would be developed, integrating principles of variable stiffness and morphing to emulate the adaptive capabilities of chelonian limbs. Prototyping and iterative testing would follow, involving the selection and optimization of materials and actuators to achieve the desired stiffness modulation and morphological changes. Experimental validation would likely entail testing the robotic limb in various simulated environments, assessing its performance in both terrestrial and aquatic locomotion scenarios. Data collected from these experiments would inform further refinements to the design and control algorithms. The methodology would culminate in a comprehensive evaluation of the robotic limb's effectiveness in mimicking chelonian adaptations and its potential for application in real-world scenarios.

Avi Cohen, And David Zarrouk. Design, Analysis and Experiments of a High-Speed Water Hovering Amphibious Robot: AmphiSTAR.

The methodology for the AmphiSTAR high-speed water-hovering amphibious robot involves initial literature review on relevant topics. Then, conceptual design is developed considering propulsion, control, and structural integrity. Numerical simulations and analytical modeling validate design parameters. Prototyping utilizes advanced materials for harsh aquatic environments. Extensive experimental testing assesses speed, stability, and energy efficiency. Data analysis informs iterative design modifications. Real-world testing evaluates performance in natural water bodies. Comprehensive evaluation highlights potential applications. Iterative refinement ensures optimization of performance. The methodology concludes with a robust assessment of the AmphiSTAR's capabilities for various tasks.

Jiefeng Sun and Jianguo Zhao. An Adaptive Walking Robot With Reconfigurable Mechanisms Using Shape Morphing Joints.

The methodology for developing the adaptive walking robot with reconfigurable mechanisms involves a literature review to understand relevant concepts. A conceptual design is formulated, considering flexibility, durability, and controllability. Prototyping

includes material selection and integration of sensors and actuators. Experimental validation assesses adaptability to various terrains and dynamic reconfiguration capabilities. Data analysis guides iterative design refinements and control algorithm adjustments. The robot's performance is comprehensively evaluated, emphasizing versatility and efficiency. The methodology ensures a systematic approach to the development and testing of the adaptive walking robot's capabilities for real-world applications.

Saad Bin Abul Kashem, Shariq Jawed Jubaer Ahmed and Uvais Qidwai. Design and Implementation of a Quadruped Amphibious Robot Using Duck Feet.

The methodology involves studying duck feet morphology and biomechanics for insights. Based on this, a conceptual design integrating biomimicry and robotics principles is formulated. Prototyping includes material selection and actuator integration for mimicking duck feet's flexibility and propulsion. Experimental validation tests the robot's locomotion in both terrestrial and aquatic environments, assessing stability and maneuverability. Data analysis guides iterative design refinements and control algorithm adjustments. The robot's performance is comprehensively evaluated, emphasizing its potential for tasks like search and rescue. This systematic approach ensures the quadruped amphibious robot's suitability for real-world applications.

B.Zhong, Y.Zhou, X.Li, M.Xu, S.Zhang Locomotion performance of the amphibious robot on various terrains and underwater with flexible flipper legs a review.

The review highlights the importance of these advancements in amphibious robotics, emphasizing the potential applications in tasks such as environmental monitoring, search and rescue missions, and underwater exploration. Challenges in control algorithms and integration of sensors for optimal performance are also discussed, contributing to the ongoing refinement of these robots. The incorporation of bioinspired designs in flipper legs further enhances the efficiency of locomotion, drawing inspiration from nature's aquatic creatures. Overall, the research underscores the promising trajectory of amphibious robots with flexible flipper legs, offering a comprehensive overview of their locomotion capabilities and potential impact on various real-world scenarios.

Junzhi Yu, Rui Ding, Qinghai Yang, Min Tan, Weibing Wang, and Jianwei Zhan On A Bio-Inspired Amphibious Robot Capable of Multimodal Motion.

The paper on a bio-inspired amphibious robot capable of multimodal motion adopts a

methodology rooted in bio-inspiration, leveraging insights from aquatic and terrestrial organisms to design a versatile robot. The approach encompasses mechanical design, sensor integration, and the development of a sophisticated control algorithm that facilitates seamless transitions between different modes of locomotion. Rigorous testing in varied environments, including water bodies and challenging terrains, validates the robot's adaptability. However, certain drawbacks are acknowledged, including the relatively high energy consumption associated with multimodal locomotion, the complexity of the control algorithm, potential maintenance issues due to the intricate mechanical design, and a limited payload capacity. These limitations point towards areas of improvement for future research, urging a balance between enhanced functionality and addressing operational challenges for practical deployment

.R. D. Maladen, Y. Ding, C. Li, and D. I. Goldman Undulatory swimming in sand: Subsurface locomotion of the sandfish lizard.

The paper on "Undulatory Swimming in Sand: Subsurface Locomotion of the Sandfish Lizard" employs a multifaceted methodology to delve into the biomechanics of the sandfish lizard's unique undulatory swimming in granular environments. Through meticulous biomechanical analysis and controlled experiments, the researchers investigate the lizard's body movements, limb coordination, and interactions with sand. The study provides valuable insights into the intricacies of subsurface locomotion, emphasizing the lizard's specialization for efficient movement in granular substrates. However, the paper acknowledges certain Advantages and also few drawbacks, including challenges in directly applying the observed locomotion to biomimetic robotics due to engineering complexities. Additionally, the study highlights the environmental constraints of the lizard's specialized adaptation to sand, raising questions about its easy adaptability to other terrains. The complexity of sand interactions is recognized as a hurdle in precisely understanding and replicating the lizard's dynamic movements. Despite these limitations, the research foundation for a deeper understanding of undulatory swimming in granular media, offering potential applications for bio-inspired robotics in specialized environments

ChenLi, PaulBumbanhowe, Haldun Komsuoglu, Daniell Goldman Sensitive Dependence Of The Motion Of A Legged Robot On Granular Media.

The study on the "Sensitive Dependence of the Motion of a Legged Robot on Granular Media" investigates the nuanced interaction between a legged robot and granular substrates. The methodology involves conducting systematic experiments to analyze

how variations in parameters such as leg morphology, gait, and substrate properties impact the robot's motion. High-speed cameras and force sensors are utilized to capture and quantify the intricate dynamics. The study systematically explores a range of conditions to understand the sensitivity of the legged robot's locomotion to granular media characteristics. Drawbacks identified in the research include challenges in precisely controlling granular media properties for consistent experimentation, potential difficulties in replicating such sensitivity in real-world applications, and the complexity of integrating these findings into broader legged robot design principles. The study sheds light on the intricate relationship between legged robots and granular environments, providing valuable insights but also highlighting challenges in translating these findings into practical and robust application.

Chapter 3

PROBLEM SPECIFICATION

3.1 Existing System

In the past two decades, significant strides have been made in the development of swimming and crawling robots, showcasing their ability to navigate through complex and unstructured environments. These robots have proven invaluable across a spectrum of applications including search and rescue operations, agricultural tasks, marine exploration, surveillance missions, and security endeavors. As technology has progressed, concerted efforts have been directed towards refining these robots, with a focus on reducing their size and weight, augmenting their speed, and minimizing energy consumption. This ongoing refinement has underscored the versatility and adaptability of these robotic systems, positioning them as indispensable tools in various sectors.

Problems with existing systems

Despite notable progress in the field of swimming and crawling robots, contemporary designs continue to grapple with significant obstacles, particularly the complexity of their propulsion mechanisms. Effective navigation across varied environments—be it water bodies, rugged terrains, or obstacle-laden landscapes—demands sophisticated propulsion systems. These systems often necessitate intricate designs incorporating multiple motors, actuators, or propulsion methods, which in turn lead to elevated developmental costs and present substantial challenges in control, maintenance, and energy consumption.

1. **Complexity of Propulsion Mechanisms:** Swimming and crawling robots require advanced propulsion systems to maneuver through diverse and often unpredictable environments. Achieving efficient and adaptable locomotion entails the integration of multiple motors and actuators, or the implementation of novel propulsion methods. This inherent complexity drives up developmental costs and complicates the control and maintenance of these robots. Sophisticated control algorithms are necessary to manage the intricate interplay of components, further complicating the design and operational processes.
2. **Energy Consumption:** The reliance on multiple motors significantly increases the energy consumption of swimming and crawling robots. Each motor demands a considerable power supply, resulting in high overall energy requirements. This heightened energy demand restricts the operational endurance and range of these robots, making them less suitable for prolonged or autonomous missions. The

challenge lies in balancing the need for powerful propulsion with the necessity for energy efficiency, a balance that is crucial for extending the operational lifespan of these robots in the field.

3. **Weight:** The complexity of propulsion systems and the inclusion of multiple motors inevitably add to the weight of swimming and crawling robots. Excess weight negatively impacts operational efficiency and maneuverability, particularly in aquatic environments where buoyancy and drag are critical considerations. The additional weight also exacerbates energy consumption issues, as more power is needed to propel heavier robots through their respective environments.
4. **Control and Maintenance Challenges:** Managing the complexity of propulsion mechanisms and multiple motors introduces significant challenges in terms of control and maintenance. Ensuring the reliability and functionality of these sophisticated systems requires advanced control algorithms and ongoing maintenance efforts. This adds to the operational burden, necessitating regular check-ups and repairs to maintain optimal performance and prevent failures.
5. **Environmental Impact:** The high energy consumption and added weight of these robots contribute to their environmental footprint. In scenarios where prolonged or autonomous operation is required, the environmental impact can be significant. This is particularly concerning given the increasing emphasis on sustainability and the need to minimize the ecological impact of technological advancements.
6. **Overcoming these limitations is essential for maximizing the potential of robotic systems in critical applications such as search and rescue, agriculture, marine exploration, surveillance, security, and reconnaissance missions.** By embracing innovation and rethinking design paradigms, the next generation of swimming and crawling robots can achieve transformative advancements. Such innovations could include the development of more efficient propulsion systems, lightweight materials, and advanced energy management technologies.

3.2 Block Diagram

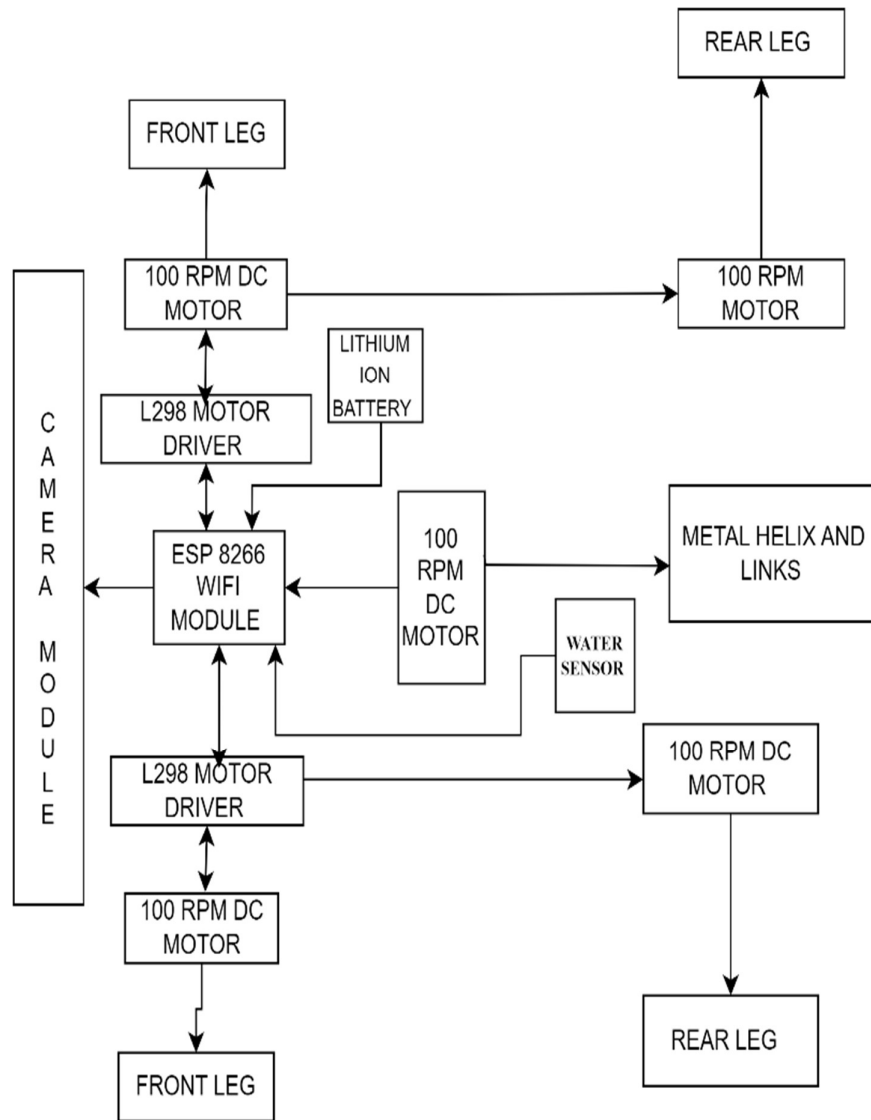


Fig.3.1 Block Diagram of system

In Figure 3.1, we can see the detailed design of the robot, especially focusing on its head section, where key components are strategically placed to ensure the robot functions well. At the center of this setup is the ESP8266, which acts as the main controller, prominently located in the head. This crucial electronic part coordinates the operation of the entire robot system.

Within the head section, two L298 motor drivers are integrated to precisely control the movement of the robot's front and rear legs using DC motors. This thoughtful design enhances the robot's mobility and flexibility, allowing for smooth and coordinated leg movements. Additionally, a camera module is housed in the head and connected to the

ESP8266 via the Wi-Fi module. This camera helps the robot monitor the terrain, providing essential visual input for navigation.

Moreover, the head section includes a connection for a saw, an important tool that adds to the robot's functionality. The saw's placement in the head is carefully considered to ensure effective use. This connection links the Arduino and the saw, enabling precise control over the saw's movements through a DC motor located in the head. A water sensor at the front controls this DC motor, ensuring the saw operates accurately and reliably.

A unique feature of the robot's propulsion system is the rotating helix, which acts as the robot's spine. This helix consists of a series of rigid plastic links that flex in one direction, similar to a bicycle chain. A single motor drives the helix, causing the linked segments to move vertically in a synchronized manner while also rotating. This motion creates a progressive sine wave that generates force at the points where the robot contacts the ground, propelling it forward efficiently.

The design of the robot's head section combines precise engineering with strategic placement of components. The central role of the ESP8266 in controlling the robot, along with the integration of motor drivers, a camera module, and a saw, highlights a careful and deliberate approach to maximizing functionality and flexibility. The rotating helix propulsion system further demonstrates innovative solutions for efficient movement. Together, these design elements show the robot's capability to navigate and perform in complex and unstructured environments, reflecting the advanced state of modern robotic technology.

In Figure 3.2, we can see more details about the saw of the robot. The saw comprises a series of inflexible acrylic links that have inherent flexibility along a single dimension, similar to a bicycle chain. The helix moves in a sine wave pattern, flexing and rotating at the same time. This intricate motion pattern allows the saw to make precise cuts and handle various tasks with accuracy.

The points of contact between the helix and the ground create a forward motion that propels the robot. This method of propulsion is both sophisticated and efficient, showcasing the advanced engineering behind the robot's design. The careful integration of these components ensures that the robot can tackle a wide range of tasks and navigate through challenging environments effectively.

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Fig3.2 Molded Plastic links

rotates, the interconnected plastic links undergo a vertical motion similar to a oscillating these links and the ground result in the exertion of force during rotation, generating a forward propulsion mechanism for every rotation of helix The utilization of inflexible molded plastic links not only imparts structural integrity to the saw but also allows for controlled and predictable motion. This design choice ensures that the saw operates with reliability and efficiency, meeting the exact requirements for crawling and floating.

The robot is controlled by the mobile app which send the commands of direction control and receives the video output from the camera module to the mobile phone.

3.3 REQUIREMENTS

Hardware Requirements

1. ESP 8266 module
2. ESP Camera module
3. L298 motor drivers
4. Water sensor
5. 100 RPM DC motors
6. Molded acrylic links
7. Copper Helix
8. Steel rods
9. Molded acrylic head
10. Hollow pipe for motor housing

Software requirements

1. Arduino IDLE
2. Blink mobile app for robot control and video reception

Connecting Power

The ESP8266, which serves as the central controller of the robot, is powered by a lead-acid battery. This lead-acid battery, with a voltage rating of 12V and a capacity of 1.3Ah, is a widely utilized rechargeable power source in numerous applications due to its reliability and cost-effectiveness. Known for its ability to provide a steady and consistent power supply, the 12V 1.3Ah lead-acid battery is an ideal choice for powering various devices and systems.

The lead-acid battery is particularly suitable for use in applications such as emergency lighting, security systems, portable tools, and small-scale renewable energy setups. Its robust construction and capability to handle deep discharges make it a dependable energy storage solution for both commercial and residential purposes. The durability and reliability of this battery type ensure long-lasting performance, which is critical for applications that require uninterrupted power supply.

In the context of the robot, this lead-acid battery is connected to the power pin of the ESP8266 module, thereby powering the entire circuit. The battery is designed to be turned on only when the robot is to be activated. This feature helps in conserving energy

and prolonging the battery life. Control of the robot is facilitated through the Blynk phone application, which provides a user-friendly interface for managing and operating the robot.

The Blynk app allows users to start the robot and control its various functions remotely, leveraging the connectivity capabilities of the ESP8266. This integration of the lead-acid battery and the Blynk app ensures that the robot can be easily operated and monitored, enhancing its usability in diverse scene.

Swimming Propulsion:

It emphasizes the significance of studying hydrodynamics and refining the swimming propulsion system for the wave-like crawling robot. This entails analyzing fluid dynamics, particularly in water, to develop efficient propulsion mechanisms. The goal is to select suitable propulsion methods, such as fins or propellers, based on effectiveness and maneuverability. Through simulations, efforts are made to minimize drag and turbulence while maximizing thrust. Material selection focuses on lightweight and durable options capable of withstanding underwater conditions. Control algorithms are integrated to ensure smooth operation, coordinating with other onboard systems. By adhering to rigorous hydrodynamic principles and optimizing the propulsion system design, the crawling robot can achieve enhanced swimming capabilities, navigating underwater environments with improved efficiency and agility.

Chapter 4

HARDWARE IMPLEMENTATION

4.1 Introduction

The hardware design and implementation for the realization of project are divided into two sections, Mechanical Design of the body and Electronic Sensor interface for the robot control. The hardware design and electronic components used are:

Mechanical Parts:

1. Molded Acrylic Head (Motor and sensor housing) (front)
2. Hollow plastic rear motor housing(back)
3. Copper Helix
4. 3D printed curved spoke wheels
5. Plastic links
6. Supporting metal rods and screws

Electrical Components:

1. ESP 8266
2. ESP camera module
3. L298 Motor Drivers
4. 100 RPM DC motors
5. Water Sensor

4.1.1 Mechanical Design

The design of the robot's body entails meticulous integration of materials and components to ensure optimal functionality and robustness. This process involves several key steps, crafting the head from acrylic, housing electronic components and provisions for wheel attachment; constructing a metal helix spine for structural integrity; utilizing plastic links to connect components in a flexible yet sturdy manner; employing a hollow cylindrical motor housing for rear wheel motors; and connecting the head to the spine and motor housing with supporting metal rods. These are conducted as follows:

1.Head Design: Imagine the robot's head as its command center, made from clear acrylic material. It's where all the magic happens – the sensors that help it see the world around it, and the motors that drive its movement. Picture it with a sleek, curvy shape,

like the head of a salamander. This design not only looks cool but also helps the robot move more efficiently. Inside the head, there are special spots designed to securely attach the front wheels, ensuring they're firmly integrated into the robot's body.

2. Spine Construction: Think of the robot's spine as its backbone, providing strength and flexibility. It's made from twisted copper metal, giving it a sturdy yet bendable structure. Just like our spine keeps us standing tall and lets us move in different directions, this metal spine helps the robot navigate through various terrains. It's a crucial part of the robot's body, ensuring it can move smoothly and adapt to different situations without breaking.

3. Link Assembly: Now, picture the robot's body held together by tough plastic links, similar to the chains on a bike. These links are cut into small rectangular pieces and screwed together, forming a chain-like structure. They're what make the robot's body flexible yet resilient, allowing it to move smoothly over rough terrain without falling apart. These links ensure that the different parts of the robot work together seamlessly, enabling it to perform its tasks effectively.

4. Tail and Motor Housing: Imagine the robot's tail as its engine room, housing the motors that power its back wheels. It's like the heart of the robot, keeping everything that makes it move safe and protected. The tail, made from a hollow plastic tube, helps the robot move forward and turn, whether it's on land or gliding through water. It ensures that the robot's movements are smooth and controlled, regardless of the terrain it encounters.

5. Wheel Design: Visualize the robot's wheels with curved spokes, made from clear acrylic material. These wheels are specially designed to help the robot move smoothly over different surfaces, such as sand, rocks, or water. They provide the traction needed for the robot to climb hills, traverse uneven terrain, and propel itself through water. The design of these wheels ensures that the robot can move efficiently and effectively, no matter where its adventures take it.

6. Wheel Attachment: Envision each set of wheels securely attached to both the head and the tail of the robot. This strategic placement ensures that the wheels remain firmly in place, providing stability and balance as the robot moves. It's like having a firm grip on both ends of a rope – it keeps the robot steady and helps it maintain its course, even in challenging conditions.

7. Structural Support: Picture metal rods connecting the head and tail of the robot, providing additional stability and support. These rods act like the robot's skeleton,

holding everything together and ensuring that it remains strong and sturdy. They provide the framework for the robot's movements, allowing it to navigate smoothly and efficiently through its environment. With this structural support in place, the robot can tackle any obstacle with confidence, exploring new terrain and gathering valuable information along the way.

Through this carefully orchestrated design, the robot achieves a harmonious balance of strength, flexibility, and functionality. Each component is thoughtfully selected and integrated to ensure the robot can navigate and operate effectively across various environments, fulfilling its intended roles in exploration and surveillance. This meticulous integration ensures the robot can operate seamlessly across varied terrains, offering reliability and precision in its tasks of exploration and surveillance.

4.1.2 Electronical Design

The electronic design of the robot are meticulously interconnected to facilitate seamless control and operation. This configuration encompasses several key elements: a central control unit, typically an ESP module, which orchestrates the robot's functions and communications; motor drivers, responsible for managing the movement of wheels or other propulsion mechanisms; sensors for collecting environmental data and facilitating navigation; and power sources, ensuring consistent energy supply to all components. These connections are made as follows:

1. **Power Supply:** The ESP module, the central control unit of the robot, is powered by rechargeable lead-acid batteries. This power supply ensures consistent and reliable energy for the robot's operations.
2. **Motor Control:** The ESP module is linked to a motor driver, which plays a crucial role in controlling the robot's movement. This connection enables precise command over the robot's locomotion.
3. **Wheel Movement:** The motor driver, receiving instructions from the ESP module, manages the movement of both the front and rear wheels. This ensures coordinated and efficient propulsion, allowing the robot to navigate various terrains smoothly.
4. **DC Motors:** High-performance DC motors are employed to drive the curved spoke wheels. These motors provide the necessary torque and speed to the wheels, facilitating the robot's versatile movement capabilities, including climbing and propelling in water.
5. **Surveillance System:** The ESP module is also connected to a camera, enabling real-time surveillance. This camera captures visual data and streams it to the control system,

providing essential feedback for monitoring and navigation.

This integrated electronic system ensures that the robot operates with high efficiency and responsiveness. The careful arrangement of power, control, and sensory components enables the robot to perform complex tasks, making it a valuable tool for exploration and surveillance in diverse environments.

4.4 Design Description

4.4.1 Mechanical Components

1. Acrylic Molded Head



Fig 4.1 Acrylic molded head outer parts

The head of the robot, as depicted in Figure 4.1, is designed using an acrylic sheet molded into a streamlined, curved shape. This design choice is inspired by the head shapes of many aquatic animals, optimizing the robot for fluid dynamics and efficient movement through water.

1. Material Selection:

Acrylic Sheet: The head is crafted from an acrylic sheet due to its waterproof properties, ensuring that the electronic components housed within are protected from moisture and water damage.

2. Streamlined Shape:

Curved Design: The head is molded into a curved, streamlined shape. This design minimizes water resistance and enhances the robot's ability to move smoothly in aquatic environments.

Biomimetic Inspiration: The shape is inspired by the natural contours of aquatic animal heads, promoting efficient movement in water and improving overall hydrodynamics.

3. Housing for Electronics:

Motors and ESP Module: The head is spacious enough to house essential electronic

components, including the motors and the ESP module, which are crucial for the robot's control and operation.

Protection: The acrylic material ensures that these components are securely enclosed and protected from environmental factors.

4. Front Wheel Attachments:

Provisions for Wheels: The head includes specific provisions for attaching two front wheels. These attachment points are designed to ensure stability and proper alignment of the wheels, facilitating smooth locomotion.

5.Connection to Helical Spine:

Link Mechanism: The head is connected to the helical spine via plastic links. These links are robust yet flexible, allowing for coordinated movement between the head and the body of the robot.

Screw Fastenings: The links are securely fastened using screws, ensuring a stable and reliable connection that can withstand various operational stresses.

6. Rear Motor Housing Connection:

Supporting Metal Rods: The head is also connected to the rear motor housing using supporting metal rods. These rods provide additional structural integrity and support.

Screw Fastenings: The metal rods are attached using screws, creating a durable and rigid framework that maintains the alignment and stability of the entire robotic structure.

7. Hollow Plastic Integration:

Tail Part: The hollow cylindrical plastic forms the tail part of the robot and serves as the housing for the rear wheels' motors.

Structural Support: The integration of hollow plastic elements contributes to the overall lightweight design while providing necessary support and housing for the motor components.

The design of the robot's head, incorporating the use of acrylic for its waterproof properties and streamlined shape, is crucial for its performance in both aquatic and terrestrial environments. The thoughtful placement of electronic components within the head, combined with the robust connections to the helical spine and rear motor housing, ensures that the robot operates with high efficiency and stability. By leveraging biomimetic design principles and high-quality materials, the robot is well-equipped to navigate diverse terrains, making it a versatile and reliable tool for environmental surveillance and deep forest exploration.

2. Rear Motor Housing



Fig4.2. Hallow Plastic Rear Motor Housing

The Hollow Plastic Rear Motor Housing in fig 4.2 is designed using cylindrical plastic pipe. This acts as the waterproof housing for the DC motors for controlling the rear wheels. This also supports the robot body to be stable while walking on the land and moving in water. The Hollow Plastic Rear Motor Housing, as depicted in Figure 4.2, is ingeniously crafted using a cylindrical plastic pipe. This component serves a dual purpose within the robot's design, functioning as both a protective enclosure for the DC motors controlling the rear wheels and as a stabilizing support structure for the entire robot body.

1. Material Selection:

Cylindrical Plastic Pipe: Chosen for its lightweight yet durable properties, providing sufficient protection for the enclosed DC motors while maintaining the overall agility of the robot.

2. Waterproof Enclosure:

Functional Design: Engineered to safeguard the DC motors from water and moisture, ensuring uninterrupted functionality even in aquatic environments.

Robust Construction: The housing's watertight design shields the motors from potential

damage, enhancing the longevity and reliability of the robot.

3. Supportive Role:

Stabilizing Structure: The rear motor housing plays a pivotal role in stabilizing the robot's body, particularly during terrestrial locomotion and aquatic movement.

Enhanced Stability: By providing a secure anchor for the rear wheels' motors, the housing contributes to the robot's balance and stability, enabling smooth navigation across various terrains.

4. Versatile Functionality:

Land and Water Adaptability: Designed to facilitate seamless transitions between land and water environments, the rear motor housing ensures the robot's adaptability and versatility in diverse operational scenarios.

Optimized Performance: By maintaining stability and protecting critical components, the housing maximizes the robot's performance potential across different terrains and conditions.

The creation of the Hollow Plastic Rear Motor Housing underscores the ingenuity and innovation inherent in the robot's design. Like a sturdy shelter for its vital organs, this component provides a safe haven for the DC motors, shielding them from external elements while ensuring the robot's stability and agility. In essence, it serves as the backbone of the robot's functionality, enabling it to confidently traverse both land and water with ease.

3.Copper Helix



Fig 4.3 Copper Helix

The design of the helix in fig 4.3 is called as single Actuator Wave design. This design

employs a helix which is powered by a motor. This helix is made using the copper metal and is molded into the shape of the helix. The Single actuator wave design is proved to be faster and simple for the movement in various terrains. The helix generates a wave-like motion along the length of the robot's body. This motion is used to the movement links in a sine wave, with peaks and troughs propagating from one end of the body to the other. As the helix rotates, the wave moves, each segment of the body bends and straightens sequentially, creating a ripple effect that propels the robot forward. The helix design depicted in Figure 4.3, known as the single Actuator Wave design, represents a remarkable innovation in robotic locomotion. This design features a helix structure powered by a motor, crafted from copper metal and molded into its distinctive shape. The single actuator wave design has been validated as both faster and simpler for navigating various terrains.

1. Wave-Like Motion:

The helix generates a wave-like motion along the length of the robot's body, resembling the undulating movement observed in aquatic creatures.

This motion propels the movement links in a sinusoidal pattern, characterized by peaks and troughs propagating from one end of the body to the other.

2. Sequential Bending and Straightening:

As the helix rotates, the wave moves along the body, causing each segment to bend and straighten sequentially. This dynamic process creates a ripple effect, wherein the bending and straightening of each segment propel the robot forward in a fluid and efficient manner.

3. Efficiency and Versatility:

The single actuator wave design offers enhanced efficiency and adaptability across diverse terrains, including rough land, wetlands, and water bodies.

By harnessing the natural mechanics of wave propagation, the design optimizes the robot's movement capabilities, allowing it to navigate complex environments with ease. The single actuator wave design represents a fascinating emulation of nature's ingenuity, drawing inspiration from the fluid dynamics of aquatic life forms. Like a creature gracefully gliding through water, the robot harnesses the power of wave-like motion to propel itself forward, navigating obstacles and terrain with remarkable agility. With each bend and straightening of its segments, the robot exhibits a mesmerizing synchronicity, reminiscent of the rhythmic undulations seen in the natural world. Through this innovative design, the robot transcends the limitations of traditional locomotion, ushering in a new era of efficiency and elegance in robotic movement.

4. 3D printed curved spoke wheels



Fig 4.4 3D printed curved spoke wheels

As shown in Fig 4.4, the curved spoke wheel mechanism is the wheel designed specifically for the robots which move on uneven terrains. These types of wheels usually have 3 or four spokes for the stability. For the present project curved spoke wheels of four spokes is used. There are two such wheels on the front and two on the rear for the movement of the robot on different terrains in a stable manner. The design parameters for these wheels are the radius of wheel, the angle between the spokes. As depicted in Figure 4.4, the curved spoke wheel mechanism is meticulously engineered for robots traversing uneven terrains. Specifically designed with stability in mind, these wheels typically feature three or four spokes. For the current project, four-spoke curved wheels have been chosen to ensure optimal stability and maneuverability.

1.Stability and Versatility:

The four-spoke design offers enhanced stability, enabling the robot to navigate various terrains with confidence and precision. By distributing the load evenly across four spokes, these wheels excel in maintaining balance and traction on uneven surfaces.

2. Placement and Functionality:

Two curved spoke wheels are positioned at the front and two at the rear of the robot,

facilitating smooth movement across diverse terrains. This configuration ensures balanced propulsion and steering capabilities, essential for navigating challenging environments effectively.

3.Design Parameters:

The design parameters for these wheels include the radius of the wheel and the angle between the spokes. These parameters are carefully calibrated to optimize performance, striking a balance between stability, agility, and terrain adaptability.

In envisioning the curved spoke wheel mechanism, we draw inspiration from nature's resilience and adaptability. Like the sturdy limbs of a creature traversing rugged terrain, these wheels offer stability and support to the robot, enabling it to conquer obstacles with ease. With each rotation, the spokes gracefully negotiate the contours of the landscape, ensuring a smooth and steady journey for the robot. As it moves across diverse terrains, the robot embodies the spirit of exploration and adventure, driven by our boundless curiosity and desire to push the boundaries of possibility. Through the innovative design of its wheels, the robot embarks on a journey of discovery, navigating the complexities of its environment with grace and determination.

5.Plastic Links

As depicted in Figure 4.5, the plastic links employed in the single actuator wave robot are meticulously engineered to combine strength with flexibility. Crafted in a rectangular shape, these links are interconnected with screws to form a continuous chain extending from the robot's head to its rear, enveloping the central copper helix. Reinforcement is provided by metal rods strategically placed throughout the assembly.

1. Strength and Flexibility:

The rectangular shape of the plastic links ensures robustness, capable of withstanding mechanical stress and environmental pressures. Despite their strength, these links retain a degree of flexibility, crucial for accommodating the dynamic movement induced by the central copper helix.

2. Chain Configuration:

Connected in a chain-like fashion, the plastic links traverse the length of the robot's body, forming a continuous and cohesive structure. This interconnected chain serves as the framework through which the helical motion is transmitted, facilitating synchronized movement across the entire robot.

3.Dynamic Oscillations:

As the central copper helix rotates, it induces vertical oscillations in the plastic

links, imparting a sinusoidal wave pattern to their motion. This rhythmic undulation mirrors the natural movements observed in aquatic organisms, enabling the robot to navigate through water with fluidity and grace.

4. Underwater Maneuverability:

The design of the single actuator wave robot, with its plastic links and central helix, is tailored for optimal performance in aquatic environments. Its ability to smoothly propel through water, reminiscent of the undulating motion of aquatic organisms, makes it particularly well-suited for underwater exploration and inspection tasks.

In embodying the essence of aquatic movement, the design of the single actuator wave robot evokes a sense of harmony and elegance. Like a creature gracefully navigating through the depths, the synchronized oscillations of its plastic links mirror the fluidity of natural underwater environments.



Fig 4.5 Plastic links (in red color)

5.Supporting rods and screws

The robot's head and the rear are connected using supporting metal rods as shown Fig 4.5 These are used for the linkage and support of the helix and link structural design. These metal rods are fixed to the front and rear end using screws.

4.2.2 ELECTRONIC COMPONENTS

1.ESP 8266

The ESP8266, depicted in the figure, functions as an independent microcontroller unit equipped with built-in Wi-Fi capabilities. Upon powering up or resetting, the ESP8266 initializes its internal hardware and peripherals. Subsequently, it scans the vicinity for available Wi-Fi networks and establishes a connection to the designated network using the provided credentials (SSID and password). Once successfully connected to the Wi-Fi network, the ESP8266 becomes capable of transmitting and receiving data packets

over the network using a variety of protocols such as TCP/IP, UDP, HTTP, MQTT, and more. Moreover, the ESP8266 features versatile peripheral control, enabling configuration of its GPIO pins to interface with external sensors, actuators, displays, and other peripheral devices. Developers have the ability to write firmware to manage these peripherals based on input from sensors or external commands received via the network. Notably, the ESP8266 supports over-the-air (OTA) firmware updates, facilitating remote firmware updates without necessitating physical access to the device hardware. This comprehensive functionality empowers developers to create flexible and robust IoT applications leveraging the capabilities of the ESP8266 microcontroller.

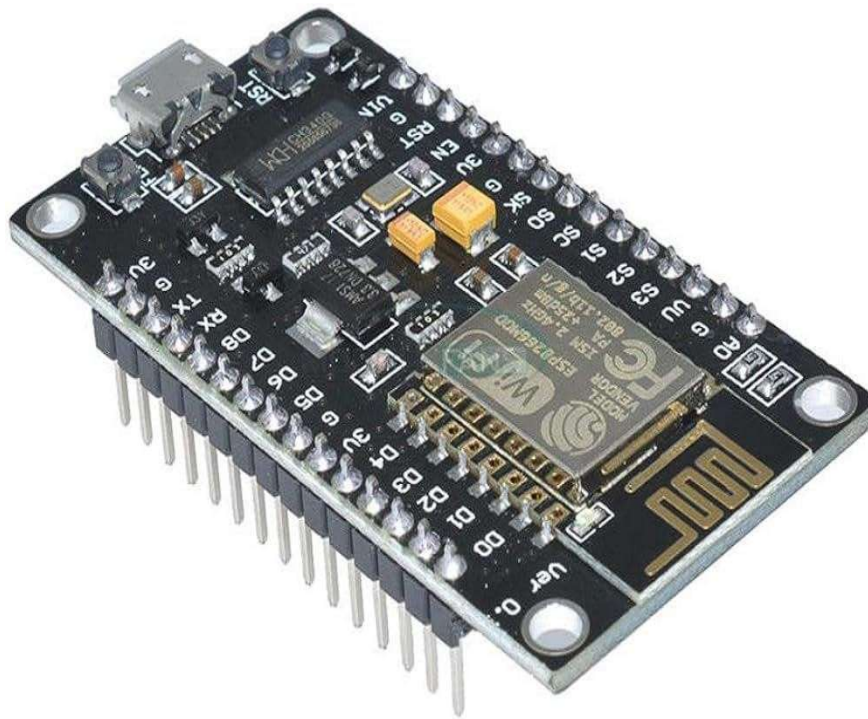


Fig 4.6 ESP 8266

Specifications:

1. Processor: L106 32-bit RISC microprocessor core based on the Tensilica Diamond Standard 106Micro running at 80 or 160 MHz.
2. Memory: 32 KiB instruction RAM
3. 32 KiB instruction cache RAM
4. 80 KiB user-data RAM
5. 16 KiB ETS system-data RAM
6. External QSPI flash: up to 16 MiB is supported (512 KiB to 4 MiB typically included).
7. IEEE 802.11 b/g/n Wi-Fi.
8. Integrated TR switch, balun, LNA, power amplifier and matching network
9. WEP or WPA/WPA2 authentication, or open networks
10. 17 GPIO pins

11. Serial Peripheral Interface Bus (SPI)
12. I²C (software implementation)
13. I²S interfaces with DMA (sharing pins with GPIO)UART on dedicated pins, plus a transmit-only UART can be enabled on GPIO2 10.10-bit ADC (successive approximation ADC)

4.2 Camera Module

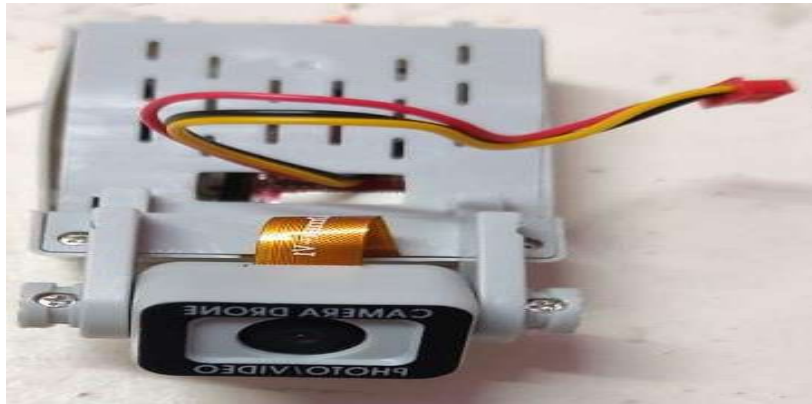


Fig 4.7 Camera module

The integration of the 200m Image Transmission Camera Module with the ESP8266 module enhances the surveillance capabilities of the Eachine EX5 seamlessly. This module is purposefully designed to capture the robot's path and transmit live video directly to a mobile phone, thereby facilitating real-time surveillance. Equipped with a high-definition camera sensor and a robust transmission system, it ensures the delivery of crisp imagery and maintains reliable connectivity over distances of up to 200 meters. When combined with the robot's control system, this setup enables precise control and stability, facilitating photography and surveillance missions both on land and in water. Moreover, its user-friendly design and intuitive interface simplify operations, making it an ideal solution for capturing stunning visuals and executing surveillance tasks with utmost ease and efficiency. This integrated system represents a significant advancement in surveillance technology, offering enhanced functionality and versatility for various applications.

Specifications:

1. Transmission Range: Up to 200 meters.
 2. Camera Resolution: High-definition.
 3. Connectivity: Seamlessly integrated with ESP 8266 module.
 4. Control Interface: Intuitive and user-friendly.
 5. Compatibility: Designed for use with the Eachine EX5 drone
- Eachine EX5 drone Features:
6. High-Definition Imaging: Captures crisp images and videos with remarkable

clarity.

7. Real-Time Transmission: Transmits live video feed to mobile phones for surveillance purposes.
8. Robust Connectivity: Ensures reliable communication over Wi-Fi signals.
9. Seamless Integration: Works in tandem with the ESP 8266 module for optimized performance.
10. Precise Control: Integrated with the drone's flight control system for stability and maneuverability.
11. Easy Operation: User-friendly interface streamlines photography and surveillance tasks.
12. Versatile Applications: Ideal for aerial photography and effective surveillance missions.

4.3 L298N Motor Driver



Fig 4.8 L298N Motor Driver

The L298N Motor Driver Module is a robust motor driver module specifically designed for driving both DC and Stepper Motors with high power requirements. Its core components include an L298 motor driver IC and a 78M05 5V regulator, ensuring stable and reliable performance. This versatile module can effectively control up to four DC motors or two DC motors with directional and speed control capabilities. In the context of the project, one L298N motor driver is allocated for controlling both the front and rear wheels of the vehicle. This configuration allows for precise control over the vehicle's movement, enabling seamless navigation across various terrains and

environments. With its ability to handle high-power motors and provide flexible control options, the L298N Motor Driver Module serves as a fundamental component in optimizing the performance and functionality of the vehicle, ensuring smooth and efficient operation in diverse scenarios.

Specifications:

1. Driver Model: L298N 2A
2. Driver Chip: Double H Bridge L298N
3. Motor Supply Voltage (Maximum): 46V
4. Motor Supply Current (Maximum): 2A
5. Logic Voltage: 5V
6. Driver Voltage: 5-35V
7. Driver Current: 2A
8. Logical Current: 0-36mA
9. Maximum Power (W): 25W
10. Current Sense for each motor
11. Heatsink for better performance
12. Power-On LED indicator

4.4 100 RPM Dc Motor



Fig 4.9 100 RPM DC Motor

The 100 RPM 12V DC Johnson Geared Motor Grade-A stands as a premier choice for applications requiring high torque and precise automation. Its standout features include a robust metal gearbox and a 6mm off-centered shaft, ensuring exceptional reliability and performance. With M3 mounting holes and a shaft length of 29mm, installation is both versatile and straightforward. This motor serves as a bridge between center shaft and side shaft gear motors, offering an ideal balance of power at a competitive price point. Despite its compact size, this motor boasts robust torque-speed characteristics,

making it suitable for a wide range of applications. Employing a mechanically commutated design, it operates efficiently on DC power supplies, ensuring consistent performance over time. The metal gearbox guarantees durability and longevity, while the shaft, fitted with metal bushes, promises extended wear resistance. Additionally, the inclusion of a hole in the shaft enhances coupling efficiency, facilitating seamless integration into diverse systems. Operating smoothly within a voltage range of 6 to 18 V DC, this motor delivers 100 RPM at 12V supply, generating an impressive torque of 12 kg-cm. Renowned for their reliability, Grade A Quality Johnson DC Motors are the preferred choice for highly capable robots, robotic platforms, and a myriad of automation tasks.

Features:

1. Grade A Quality branded Johnson DC Motor.
2. Sturdy construction ensures durability.
3. Shaft equipped with metal bushes for enhanced longevity.
4. High-quality gears ensure smooth operation.
5. Shaft features a hole for improved coupling efficiency.

4.5 Rain Water Sensor

A rain sensor, also known as a rain detector or rain gauge, plays a crucial role in meteorological and automated systems by detecting rainfall and triggering appropriate responses or alerts. These devices operate by measuring the presence and intensity of rainwater, employing various detection mechanisms to accomplish this task. Common methods include conductivity, capacitance, or optical properties detection.

Conductivity-based sensors utilize electrodes to measure the electrical conductivity of rainwater. As moisture levels increase, so does the electrical conductivity, allowing these sensors to detect the presence of rain.

Capacitance sensors work by assessing changes in the electrical capacitance between two plates as rainwater accumulates. This change in capacitance serves as an indicator of rainfall intensity and duration.

On the other hand, optical sensors rely on alterations in light reflection or refraction caused by rain droplets. By detecting these changes, optical sensors can determine the presence and intensity of rainfall.

Through accurate detection of rainfall, rain sensors enable automated systems to respond promptly to changing weather conditions. This functionality facilitates a variety of tasks, including controlling irrigation systems based on actual rainfall, activating

vehicle windshield wipers when rain is detected, or implementing flood prevention measures by alerting authorities to potential flooding risks.

Overall, rain sensors play a vital role in enhancing efficiency and safety in various applications, ensuring that automated systems can adapt to and mitigate the effects of changing weather conditions effectively.

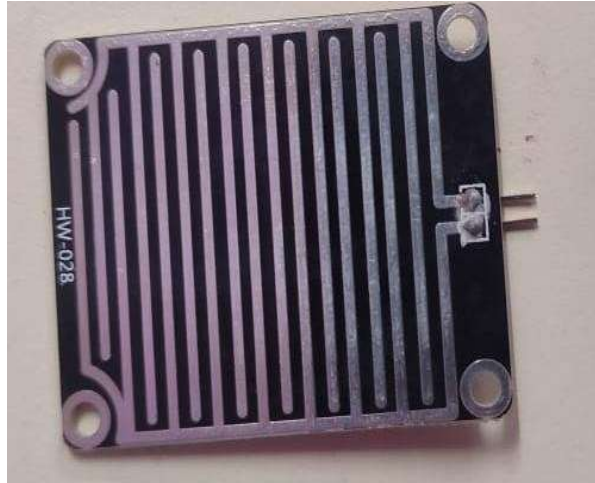


Fig 4.10 Rain Water Sensor

PIN configurations:

1. Pin1 (VCC): It is a 5V DC pin
2. Pin2 (GND): it is a GND (ground) pin
3. Pin3 (DO): It is a low/ high output pin
4. Pin4 (AO): It is an analog output pin

Specifications:

1. This sensor module uses good quality of double-sided material.
2. Anti-conductivity & oxidation with long time use
3. The area of this sensor includes 5cm x 4cm and can be built with a nickel plate on the side
4. The sensitivity can be adjusted by a potentiometer
5. The required voltage is 5V
6. The size of the small PCB is 3.2cm x 1.4cm
7. For easy installation, it uses bolt holes
8. It uses an LM393 comparator with wide voltage

Chapter 5

SOFTWARE IMPLEMENTATION

The software implementation of our project represents a pivotal aspect of its overall functionality, harnessing the power of advanced technologies to drive seamless control and interaction. Leveraging the Arduino IDE, along with the Blynk and RCFV Pro mobile applications, our software architecture epitomizes innovation and efficiency in the realm of robotics and automation.

Arduino IDE stands as the cornerstone of our software development, providing a versatile and user-friendly platform for programming the intricate functionalities of our robotic system. With its intuitive interface and extensive library of pre-built functions, Arduino IDE empowers us to translate complex algorithms and logic into executable code, ensuring precise control and operation of our robotic hardware.

Complementing the capabilities of Arduino IDE are the Blynk and RCFV Pro mobile applications, which serve as indispensable tools for remote monitoring and control. Through seamless integration with Arduino-based devices, Blynk enables real-time communication and data visualization, allowing users to remotely monitor and interact with our robotic system from anywhere in the world. Similarly, the RCFV Pro mobile application provides an intuitive interface for live video monitoring for surveillance purposes, enhancing user experience and customization capabilities.

The integration represents a synergistic approach to software implementation, driving innovation and efficiency in our project.

5.1 ARDUINO IDE

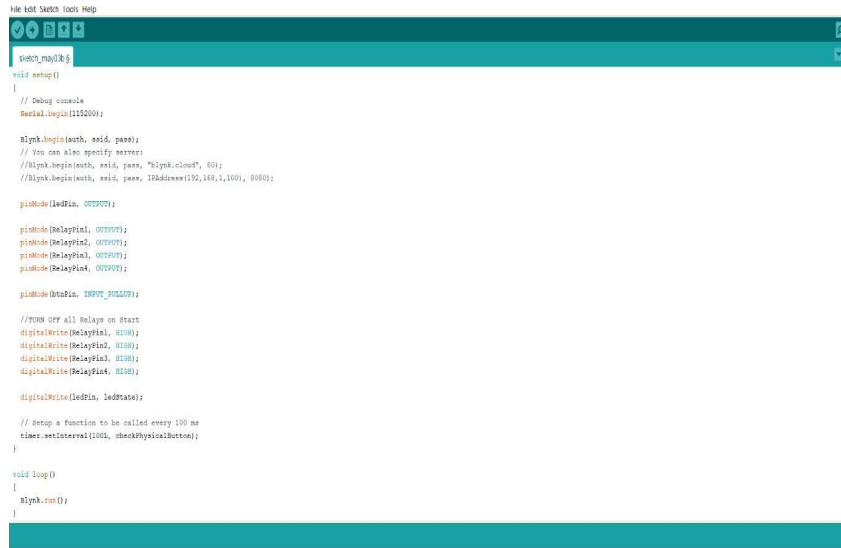


Fig 5.1 Arduino IDE

Arduino IDE (Integrated Development Environment) serves as a pivotal tool in the realm of embedded systems and microcontroller programming, offering a comprehensive platform for developing and deploying code efficiently. Its user-friendly interface and robust feature set make it a preferred choice among hobbyists, students, and professionals alike. In this essay, we will explore the Arduino IDE and delineate the steps to create a program for the ESP8266 microcontroller, a versatile component widely utilized in IoT (Internet of Things) applications.

Arduino IDE provides a cohesive environment for writing, compiling, and uploading code to a plethora of microcontroller platforms, including the popular Arduino boards and ESP8266 modules. Its intuitive interface simplifies the coding process, allowing users to focus on developing their projects without grappling with complex setup procedures or arcane syntax.

Programming an ESP8266 module using Arduino IDE involves a series of systematic steps to ensure smooth development and deployment of code. Firstly, users must ensure the installation of board definitions and libraries by navigating to the "Board Manager" within the "Tools" menu. Here, the appropriate ESP8266 board variant is selected and installed, followed by choosing the ESP8266 board from the "Tools > Board" menu, alongside specifying the corresponding processor and port settings. Once the board configuration is established, the next step is writing the code. Arduino IDE provides support for a variant of the C++ programming language, supplemented by a wealth of libraries and examples tailored to various hardware peripherals and functionalities. This phase involves defining variables, setting up pins and peripherals, implementing logic and algorithms, and managing communication protocols such as Wi-Fi or MQTT. Throughout the programming process, users can leverage available resources such as Arduino IDE's documentation, forums, and online communities for guidance, issue troubleshooting, and collaboration with fellow enthusiasts.

To ensure the correctness of the code, the "Verify" button is utilized to compile it. Arduino IDE's built-in compiler flags syntax errors and semantic inconsistencies, aiding in debugging and maintaining code integrity.

Finally, the program is uploaded to the ESP8266 module by clicking the "Upload" button. Users select the appropriate communication port for flashing the program onto the module, thereby completing the programming process. These structured steps, when followed diligently, empower users to effectively utilize the

ESP8266 microcontroller and embark on ventures in embedded systems and IoT with confidence and precision.

Arduino IDE is equipped with a diverse array of features meticulously crafted to cater to the needs of both novice and seasoned developers within the domain of embedded systems and microcontroller programming. Its user-friendly interface streamlines the coding process, ensuring a seamless and intuitive development journey. Among its notable attributes is a versatile text editor boasting features like syntax highlighting and auto-completion, which greatly enhance the efficiency of code creation and refinement. The inclusion of a built-in compiler guarantees swift and accurate code verification, effectively pinpointing errors and inconsistencies to facilitate seamless debugging. Moreover, Arduino IDE's extensive library repository offers access to a rich assortment of pre-written code snippets and examples, empowering users to accelerate development while fully leveraging the capabilities of supported hardware peripherals. Furthermore, its broad compatibility across various microcontroller platforms, coupled with comprehensive documentation, solidifies its status as an indispensable tool for crafting innovative projects and delving into the intricate realms of electronics and automation.

5.2 Blynk Application

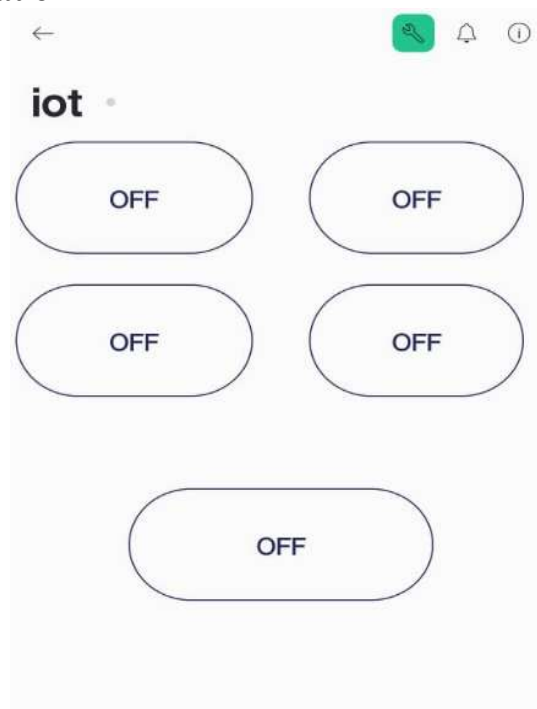


Fig 5.2 Blynk Application Interface

Blynk is a revolutionary mobile application that empowers users to effortlessly control and monitor their Internet of Things (IoT) devices from anywhere in the world

with just a few taps on their smartphone or tablet. Founded in 2015, Blynk has rapidly emerged as a leading platform for IoT enthusiasts, hobbyists, and professionals seeking to create intuitive and interactive interfaces for their connected devices. Blynk provides a user-friendly interface for building customizable IoT projects without the need for extensive programming knowledge or hardware expertise. With Blynk, users can seamlessly connect a wide range of hardware platforms, including Arduino, Raspberry Pi, ESP8266, and more, to their mobile devices via Wi-Fi, Bluetooth, or cellular networks.

The hallmark feature of the Blynk app is its drag-and-drop interface, which allows users to effortlessly design custom control panels, or "dashboards," for their IoT projects. Users can choose from a variety of widgets such as buttons, sliders, gauges, graphs, and displays, and arrange them on the dashboard to create intuitive interfaces tailored to their specific needs.

One of the key advantages of Blynk is its extensive library of pre-built widgets and templates, which enable users to quickly create sophisticated control panels for their IoT devices with minimal effort. Whether monitoring temperature and humidity sensors, controlling lights and appliances, or receiving alerts and notifications, Blynk provides a comprehensive suite of tools to bring IoT projects to life.

Furthermore, Blynk offers a wide range of connectivity options, allowing users to communicate with their IoT devices via Wi-Fi, Bluetooth, Ethernet, GSM, or even LoRaWAN. This versatility ensures compatibility with virtually any IoT project, regardless of its communication requirements or physical location.

In addition to its intuitive interface and extensive feature set, Blynk also provides robust security measures to protect users' data and privacy. All communication between the Blynk app and IoT devices is encrypted using industry-standard protocols, ensuring that sensitive information remains secure at all times.

Another notable feature of Blynk is its cloud-based infrastructure, which enables users to access and control their IoT devices from anywhere in the world, as long as they have an internet connection. This remote access capability allows users to monitor and manage their IoT projects in real-time, regardless of their physical location.

Moreover, Blynk offers a range of advanced features and capabilities for more experienced users, including support for scripting and automation, data logging and analytics, and integration with third-party services and platforms such as IFTTT and Zapier. These features enable users to create complex, multi-functional IoT projects

with ease, while also providing opportunities for integration with other systems.

5.3RCFPV PRO Application

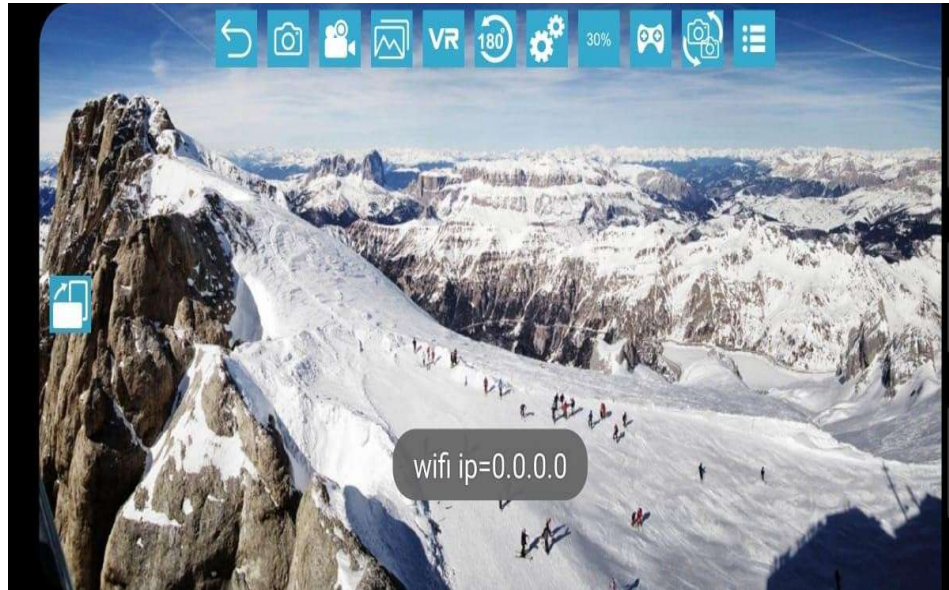


Fig 5.3 RCFPV PRO App Interface

RC FPV, developed by cooingdv, represents a cutting-edge Android application engineered to elevate the control and monitoring capabilities of robots and drones. Leveraging wireless Wi-Fi HD image transmission technology, the app establishes a seamless connection between the user's smartphone and the robotic device. This connectivity is pivotal for real-time control and video streaming, empowering users to navigate their quadcopters with precision through an intuitive joystick interface. Moreover, the application furnishes users with real-time video feeds from the aircraft's camera, facilitating the capture and management of high-quality photos and videos directly from their devices.

Beyond its foundational functionalities, RC FPV incorporates several innovative features tailored to cater to both novice and seasoned quadcopter enthusiasts. Gravity sensing technology facilitates intuitive control, enabling users to maneuver the device by tilting their smartphones. The trajectory drawing functionality further enriches the user experience by enabling the creation of automated flight paths, ensuring consistent and fluid flight routines. To offer a more immersive experience, RC FPV includes a VR mode that, when coupled with a compatible VR headset, delivers a first-person view from the quadcopter, enhancing the flight experience with heightened engagement and realism.

RC FPV's comprehensive suite of tools positions it as an indispensable asset for

a diverse array of applications. For recreational users, the app facilitates casual flying and the capture of breathtaking aerial views. Additionally, it caters to photographers and videographers by furnishing advanced media capture and management capabilities, enabling the creation of unique aerial shots for their portfolios. Moreover, FPV racing enthusiasts can leverage the real-time streaming and VR mode functionalities to augment their competitive edge. RC FPV emerges as a premier application in the realm of quadcopter technology, offering robust and versatile functionalities meticulously tailored to enhance overall surveillance and elevate the user experience to new heights.

Chapter 6

INTEGRATION AND TESTING

6.1 Integrating the ESP 8266 and ARDUINO IDE

Turn on the robot using the switch provided on the head of the robot. This turns on the ESP modules and the camera. The ESP module is used for the control of the robot directions and movement and the Camera live video transmission to the mobile applications.

The motor drivers and the camera module are connected to the GPIO pins of the ESP 8266 module. Additional resistors and capacitors may need to form the part of the circuit for the sensors. The robot control and the camera are controlled using Blynk mobile application.

To activate the robot, use the switch provided on the head of the robot. Engaging this switch powers on the ESP8266 modules and the camera. The ESP8266 module serves as the primary control unit, managing the robot's directional movements and transmitting live video feed from the camera to the mobile application.

The motor drivers and the camera module are connected to the GPIO pins of the ESP8266 module. This setup allows the ESP8266 to precisely control the movement of the robot by regulating the motor drivers. Additionally, the camera module captures real-time video, which is then transmitted to the user's mobile device via the ESP8266's Wi-Fi capabilities. To ensure proper operation and signal conditioning, additional resistors and capacitors may be integrated into the circuit to support the sensors.

The entire control system, including the robot's movement and camera functionality, is managed through the Blynk mobile application. This app provides an intuitive interface for users to remotely control the robot, monitor its movements, and view live video feeds, enhancing the user experience and operational efficiency

6.2 Testing

The fig 6.2.1 represents the Arduino Ide with code for the integration of the hardware and the Software applications. The testing phase is critical to ensure that all components of the robot function correctly and in unison. The following steps outline the testing procedure

1. Initial Power-Up:

- i. Turn on the robot using the switch on the head.
- ii. Verified that the ESP8266 modules and the camera are powered up.

2. Connectivity Check:

- i. Ensured that the ESP8266 is successfully connected to the Wi-Fi network.
- ii. Confirmed that the Blynk mobile application is connected to the ESP8266 module.

3. Motor Driver Testing:

- i. The GPIO connections to the motor drivers are tested
- ii. Each motor's response to control commands from the Blynk app, ensuring smooth and coordinated movements of the robot's legs using the blynk application shown in Fig 6.2.

4. Camera Functionality:

- i. Verified that the camera module is properly capturing and transmitting live video to the mobile application RCFPV pro app in fig 6.2.3
- ii. Checked the video quality and ensure if there is minimal lag in the transmission.

5. Sensor Integration:

- i. Tested the additional resistors and capacitors in the circuit to ensure they are providing the necessary signal conditioning for the sensors.
- ii. Validated the accuracy and responsiveness of any sensors connected to the robot

6. Full System Test:

- i. Performed a comprehensive test by using the Blynk mobile application to control the robot's movement and monitor the live video feed simultaneously.
- ii. Executed a series of maneuvers to ensure the robot can navigate as expected.

7. Reliability and Endurance:

- i. Conducted prolonged operation tests to evaluate the battery life and the
- ii. robot's performance over time.
- iii. Monitor the system for any signs of overheating or malfunction.

By thoroughly testing each component and the integrated system, we ensure that the robot operates reliably and meets the project's specifications. This systematic approach to testing will help identify and rectify any issues, ensuring a robust and efficient robotic

system

```
File Edit Sketch Tools Help
sketch_main[28]
void setup()
{
  // Debug console
  Serial.begin(115200);

  Blynk.begin(auth, ssid, pass);
  // You can also specify server:
  //Blynk.begin(auth, ssid, pass, "blynk.cloud", 80);
  //Blynk.begin(auth, ssid, pass, IPAddress(192,168,1,100), 8080);

  pinMode(LED_PIN, OUTPUT);

  pinMode(RelayPin1, OUTPUT);
  pinMode(RelayPin2, OUTPUT);
  pinMode(RelayPin3, OUTPUT);
  pinMode(RelayPin4, OUTPUT);

  pinMode(StopPin, INPUT_PULLUP);

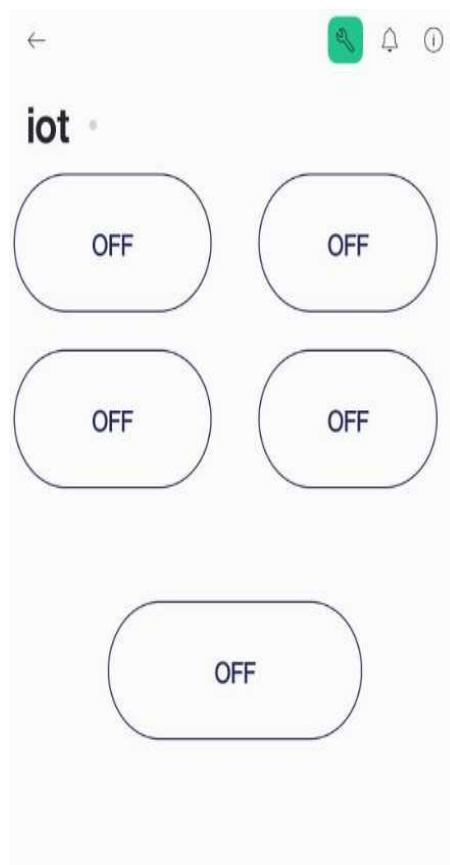
  //TURN OFF all Relays on Start
  digitalWrite(RelayPin1, HIGH);
  digitalWrite(RelayPin2, HIGH);
  digitalWrite(RelayPin3, HIGH);
  digitalWrite(RelayPin4, HIGH);

  digitalWrite(LED_PIN, LOW);

  // Setup a function to be called every 100 ms
  timer.setInterval(100L, checkPhysicalButton);
}

void loop()
{
  Blynk.run();
}
```

Fig 6.1 program



The fig 6.2.The Interface of Blynk Application

Chapter7

RESULT

The project results and the hardware kit of the amphibious robot are shown in this chapter by accomplishing the objectives.

7.1 The Robot

Upon successfully interfacing all the modules according to the proposed system and connecting them to the power supply, the complete hardware setup is illustrated in Figure 7.1. This figure serves as a visual representation of the comprehensive hardware configuration of the project, showcasing all essential components seamlessly integrated to function as a cohesive unit. From the central processing unit to the various sensors, actuators, and communication modules, every element is meticulously arranged and interconnected to ensure optimal functionality and performance of the system. The illustration provides a clear overview of the entire hardware infrastructure, highlighting the intricate interplay between different components and their collective contribution to the overall system operation. Through this integrated approach, the project achieves its objectives effectively, offering a robust and reliable solution for its intended application.

7.2 Results

We've developed a groundbreaking amphibious robot featuring a transformable flipper-leg propulsion system, drawing inspiration from biological models. This innovative design, based on the single actuator wave mechanism, is equipped with ESP8266 and camera modules. Specifically crafted for monitoring and deep forest exploration, this robot seamlessly moves across various terrains, showcasing its versatility and adaptability.

Extensive testing of the robot's movement and camera control functionalities has yielded impressive results. Accessible via a mobile application, the data provides real-time feedback on wheel status, robot direction, and live camera footage. The robot dynamically adjusts its direction, helix mechanism, and camera operations to suit varying terrains and commands, ensuring efficient and precise performance in any environment.

Figure 7.1 showcases a significant milestone: the robot's successful locomotion on water surfaces. This achievement is made possible by the synchronized movement of the helix mechanism and curved spoke wheels. Their seamless integration enables the robot to navigate aquatic environments with exceptional efficiency and

precision, marking a significant advancement in its capabilities and opening up new possibilities for aquatic exploration and surveillance.

Similarly, Figure 7.2 demonstrates the robot's prowess on terrestrial terrain. The unique design of the curved spoke wheels ensures traction and facilitates smooth movement on solid ground. This successful demonstration underscores the robot's adaptability and efficiency, highlighting its ability to traverse diverse environments with agility and proficiency, making it an indispensable tool for various applications, from environmental monitoring to search and rescue missions.



Fig 7.1 The robot movement on water surface



Fig 7.2 The robot movement on the land

In two scenarios, the integration and synchronization of the various components within the robot's design exemplify a meticulous engineering endeavor aimed at achieving optimal performance across different operational contexts. Collectively, these achievements reflect the sophisticated integration and precise synchronization of the robot's components, resulting in a highly versatile and efficient multifunctional system. The i'novative design and operational success of the robot mark a significant advancement in robotic technology, paving the way for future developments and applications in a wide array of sectors. This progress heralds a new era of exploration and practical utility, showcasing the transformative potential of advanced robotic systems in addressing complex environmental and operational challenges.

Chapter 8

CONCLUSION

8.1 Conclusion

In conclusion, the wave-like crawling robot stands as a remarkable achievement in the realm of robotic locomotion, signaling a significant milestone in the field. Its innovative design and unparalleled versatility empower it to traverse a diverse array of terrains with remarkable efficiency, from the rugged landscapes of deep forests to wetlands, sandy areas, and even ponds. This exceptional capability renders it ideally suited for applications in deep forest surveillance, where navigating through challenging environments is essential.

The pioneering technology embodied in this robot holds immense promise for revolutionizing various industries. Its exceptional adaptability and performance not only expand the scope of robotic and automated systems but also pave the way for groundbreaking advancements in fields ranging from environmental monitoring to search and rescue operations. As a testament to human ingenuity, the wave-like crawling robot serves as a beacon of innovation, exemplifying the potential for future advancements in robotics and automation. Its development heralds a new era of technological progress and offers practical solutions to complex real-world challenges, underscoring the transformative impact of cutting-edge robotics in addressing society's evolving needs and aspirations.

8.2 Future Scope

The application potential of amphibious robots equipped with single actuator wave mechanisms and curved spoke wheels extends across various domains, particularly in the realm of environmental surveillance rovers. Their unique design and versatile capabilities offer a holistic solution for monitoring and exploring diverse terrains, including rough land, wetlands, and water bodies. This adaptability makes them exceptionally suited for applications requiring robust and efficient mobility in challenging environmental conditions.

These advanced robots hold significant promise for enhancing environmental surveillance efforts. Their ability to navigate and traverse challenging landscapes ensures comprehensive monitoring and data collection, which is crucial for ecological research, conservation initiatives, and disaster management. By leveraging their sophisticated locomotion capabilities, these amphibious robots represent a significant leap forward in the deployment of automated systems for environmental

monitoring. Their contribution underscores the ongoing evolution and impact of robotic technologies in addressing complex environmental challenges, marking a pivotal step towards achieving sustainable and resilient ecosystems. With their ability to access remote and difficult-to-reach areas, these robots empower researchers and conservationists to gather valuable data and insights, ultimately contributing to informed decision-making and effective management of natural resources.

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APPENDIX

THE CODE

```
#define BLYNK_TEMPLATE_ID "xxxxxxxxx"
#define BLYNK_DEVICE_NAME "Testing Template"
#define BLYNK_AUTH_TOKEN "xxxxxxxxx"

// Comment this out to disable prints and save space
#define BLYNK_PRINT Serial

#include <ESP8266WiFi.h>
#include <BlynkSimpleEsp8266.h>

char auth[] = BLYNK_AUTH_TOKEN;
char ssid[] = "xxxxxxxxx";
char pass[] = "xxxxxxxxx";

// Set your LED and physical button pins here
const int ledPin1 = 2;
const int ledPin2 = 2;
const int ledPin3 = 2;
const int btnPin1 = 1;
const int btnPin2 = 3;
const int btnPin3 = 13;
const int btnPin4 = 15;

BlynkTimer timer;

void checkPhysicalButton();

int ledState = LOW;
int btnState = HIGH;

int RelayPin1 = 4;
int RelayPin2 = 5;
int RelayPin3 = 12;
int RelayPin4 = 14;

// Every time we connect to the cloud...
BLYNK_CONNECTED() {
  // Request the latest state from the server
  Blynk.syncVirtual(V0);
  Blynk.syncVirtual(V1);
  Blynk.syncVirtual(V2);
  Blynk.syncVirtual(V3);
}

BLYNK_WRITE(V0) {
  int pinValue = param.asInt();
```

```

    digitalWrite(RelayPin1, pinValue);
}
BLYNK_WRITE(V1) {
    int pinValue = param.asInt();
    digitalWrite(RelayPin2, pinValue);
}
BLYNK_WRITE(V2) {
    int pinValue = param.asInt();
    digitalWrite(RelayPin3, pinValue);
}
BLYNK_WRITE(V3) {
    int pinValue = param.asInt();
    digitalWrite(RelayPin4, pinValue);
}

void checkPhysicalButton()
{
    if (digitalRead(btnPin) == LOW) {
        // btnState is used to avoid sequential toggles
        if (btnState != LOW) {

            // Toggle LED state
            ledState = !ledState;
            digitalWrite(ledPin, ledState);

            // Update Button Widget
            Blynk.virtualWrite(V2, ledState);
        }
        btnState = LOW;
    } else {
        btnState = HIGH;
    }
}

void setup()
{
    // Debug console
    Serial.begin(115200);

    Blynk.begin(auth, ssid, pass);
    // You can also specify server:
    //Blynk.begin(auth, ssid, pass, "blynk.cloud", 80);
    //Blynk.begin(auth, ssid, pass, IPAddress(192,168,1,100), 8080);

    pinMode(ledPin, OUTPUT);

    pinMode(RelayPin1, OUTPUT);
    pinMode(RelayPin2, OUTPUT);
    pinMode(RelayPin3, OUTPUT);
    pinMode(RelayPin4, OUTPUT);
}

```

```
pinMode(btnPin, INPUT_PULLUP);

//TURN OFF all Relays on Start
digitalWrite(RelayPin1, HIGH);
digitalWrite(RelayPin2, HIGH);
digitalWrite(RelayPin3, HIGH);
digitalWrite(RelayPin4, HIGH);

digitalWrite(ledPin, ledState);

// Setup a function to be called every 100 ms
timer.setInterval(100L, checkPhysicalButton);
}

void loop()
{
  Blynk.run();
}
```